

STP Attacks:

- o Inject a BPDU claiming to be root to cause spanning-tree recalculations.
- o Inject a BPDU claiming to be root and creating paths in the whole network.
- o Rogue switch configured with lower BID become Root Bridge on the network.
- o This could cause inefficient traffic flow & traffic will go through the rogue switch.
- o Network Attackers can launch different types of attacks on Spanning Tree Protocol.
- o Cisco Switches have different features & mitigation for protection against STP attacks.
- o Root Guard, BPDU Guard and BPDU Filter are some features available for protection.
- o Before launching Spanning Tree Protocols (STP) attack on Cisco Switch, let us check SW.
- o The Cisco Switch is the root and all the interfaces are designated and forwarding status.

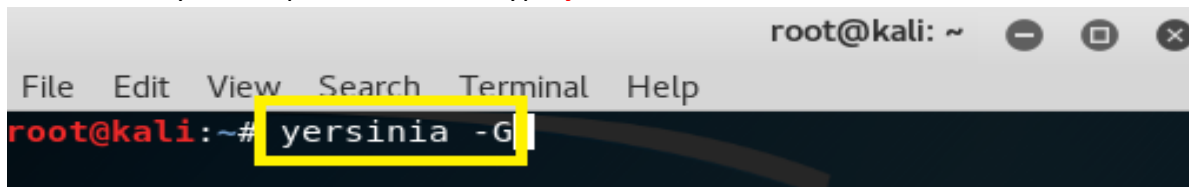
```
SW#show spanning-tree vlan 1
```

```
VLAN0001
Spanning tree enabled protocol ieee
Root ID    Priority    32769
           Address    aabb.cc00.2000
           This bridge is the root
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

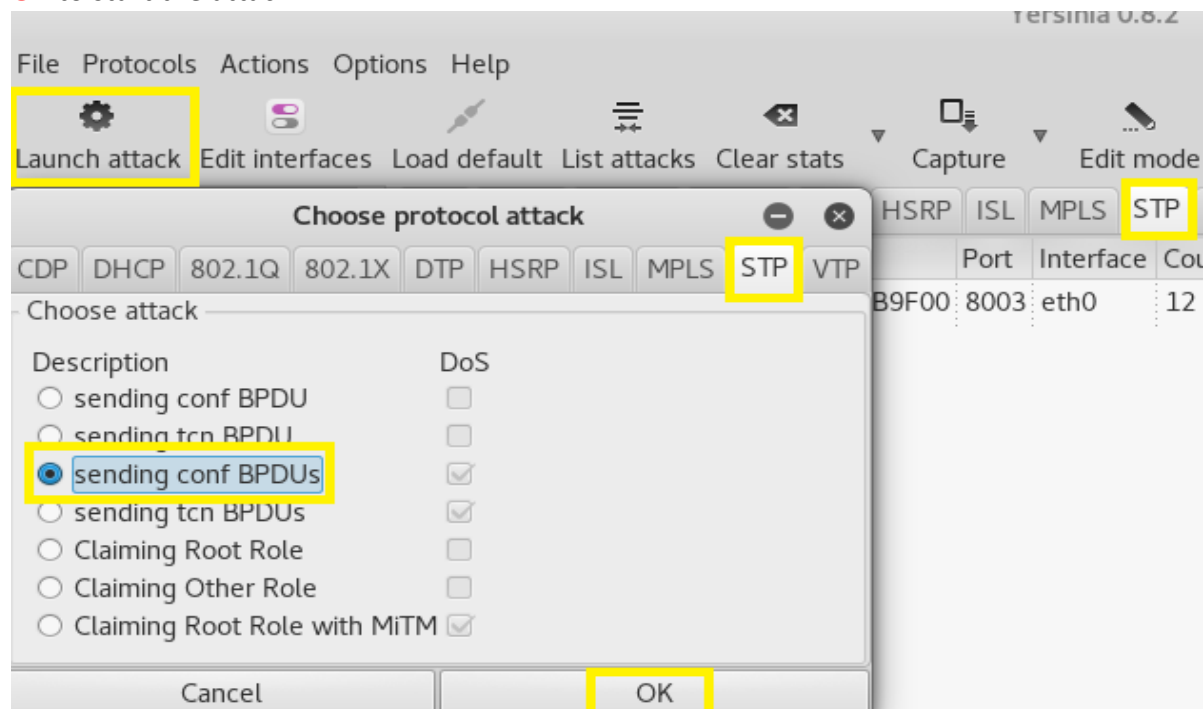
Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
           Address    aabb.cc00.2000
           Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
           Aging Time 300 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Et0/0	Desg	FWD	100	128.1	P2p
Et0/1	Desg	FWD	100	128.2	P2p
Et0/3	Desg	FWD	100	128.4	P2p

In **Attacker** System, open Terminal in type **yersinia -G**.



Click on **STP** tab go to main menu click on **Launch attack** click on **sending conf BPDUs** and click **OK** to start the attack.



Now check the spanning tree in **SW** it is not anymore root switch and the port connected to Attacker system become Root Port.

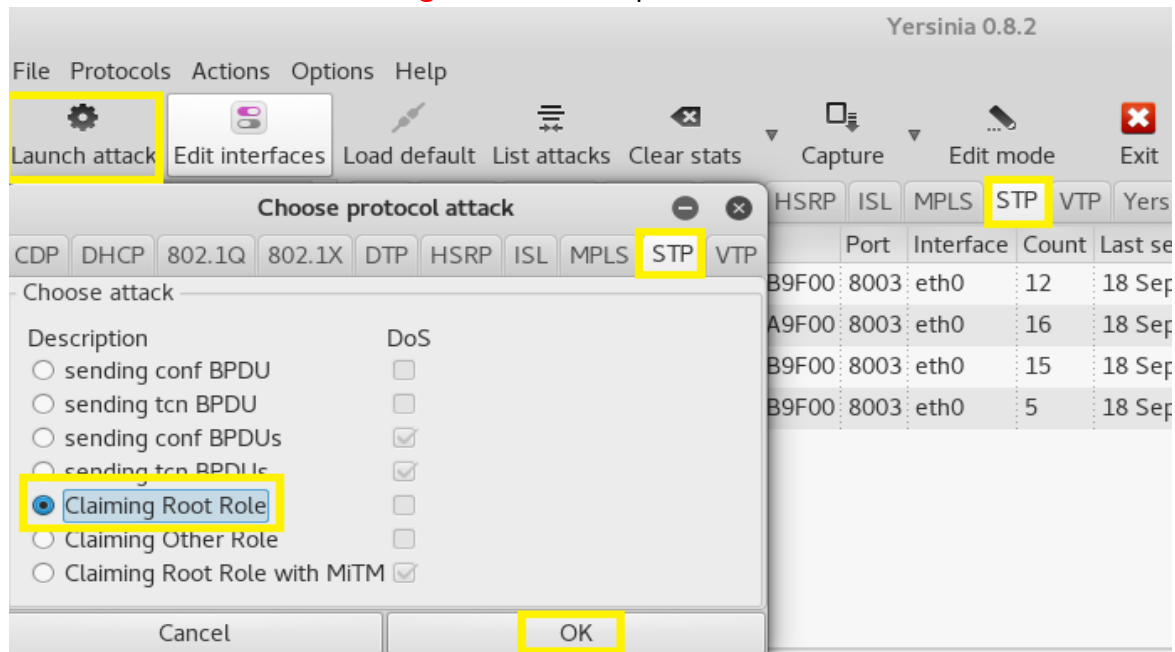
SW#show spanning-tree vlan 1

```
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    0
             Address     a895.a838.31de
             Cost        100
             Port        1 (Ethernet0/0)
             Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec

  Bridge ID  Priority    32769 (priority 32768 sys-id-ext 1)
             Address     aabb.cc00.2000
             Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec
             Aging Time   15 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Et0/0	Root	FWD	100	128.1	P2p
Et0/1	Desg	FWD	100	128.2	P2p
Et0/3	Desg	FWD	100	128.4	P2p

Another STP attack to get **Root Port Role**. Click on **STP** tab go to main menu click on **Launch attack** this time choose **Claiming Root Role** and press **OK** to start the attack.



Now check back the switch is not anymore Root switch also the port is the Root.

SW#show spanning-tree vlan 1

```
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    0
             Address     a895.a838.31de
             Cost        100
             Port        1 (Ethernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     aabb.cc00.2000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time   15 sec
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Et0/0	Root	FWD	100	128.1	P2p
Et0/1	Desg	FWD	100	128.2	P2p
Et0/3	Desg	FWD	100	128.4	P2p

BPDU Guard:

- o BPDU Guard is features that helps you protect your spanning-tree topology.
- o BPDU Guard feature can be enabled globally at Global configuration mode.
- o BPDU Guard feature can be enabled per Interface at Interface configuration mode.
- o When a BPDU Guard enabled port, receive BPDU from the connected device.
- o BPDU Guard disables the port and the port state is changed to errdisable state.

Root Guard:

- o Root Guard protects STP topology attack of replacing Root Bridge with a rogue Root.
- o Root Guard feature enabled switch port receives a superior BPDU from a rogue switch.
- o Root Guard feature change the state of the port into root-inconsistent state.
- o Root Guard feature enforcing the position of original Root Bridge in Cisco switches.
- o Port state root-inconsistent state, no user data is sent via that Cisco Switch port.
- o After the flow of superior BPDUs is stopped, port state change back to forwarding state.
- o Root Guard feature enabled on ports that is connected to other switches not root switch.

Mitigation of STP Attack
SW(config)#interface ethernet0/0
SW(config-if)#spanning-tree portfast edge
SW(config)#spanning-tree portfast edge default
SW(config)#spanning-tree portfast edge bpduguard default
SW(config)# interface ethernet0/0
SW(config-if)#spanning-tree bpduguard enable
SW(config)# interface ethernet0/0
SW(config-if) #shutdown
SW(config-if)# no shutdown
SW(config)#errdisable recovery cause bpduguard
SW(config)#errdisable recovery interval 30
SW# show errdisable recovery
SW# show spanning-tree summary
SW# show spanning-tree interface ethernet0/0 detail
SW# show interfaces status
SW# show interfaces status err-disabled

Configure Root Guard
SW(config)#interface ethernet0/0
SW(config-if)#spanning-tree guard root
SW# show spanning-tree detail
SW# show spanning-tree inconsistentports
SW# show spanning-tree interface ethernet0/0 detail